



Generate baseline

Design the ski's unweighted side profile: a camber pocket between the rocker contact points and fore and aft rockers out to the tips -- optionally shaped by the ski's own EI.

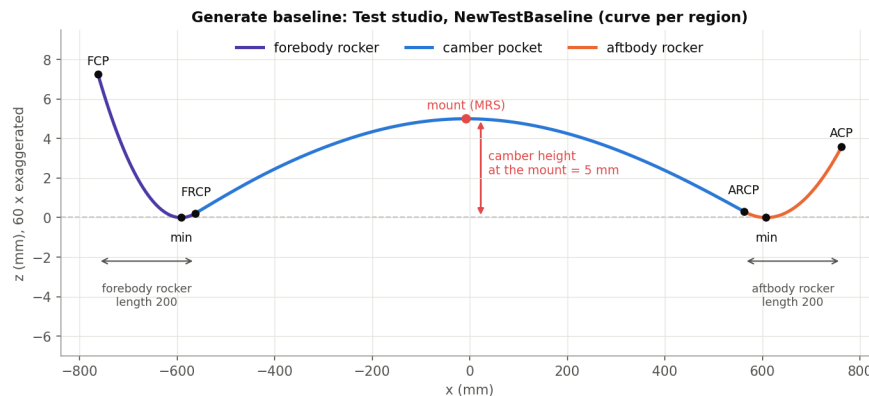


What it does

A baseline is a handful of targets -- camber height, rocker lengths, tip heights -- that must meet in one smooth curve. Generate baseline builds it from those numbers, levels it on its rocker low points, and can shape the camber as the ski actually bends.

Use it when

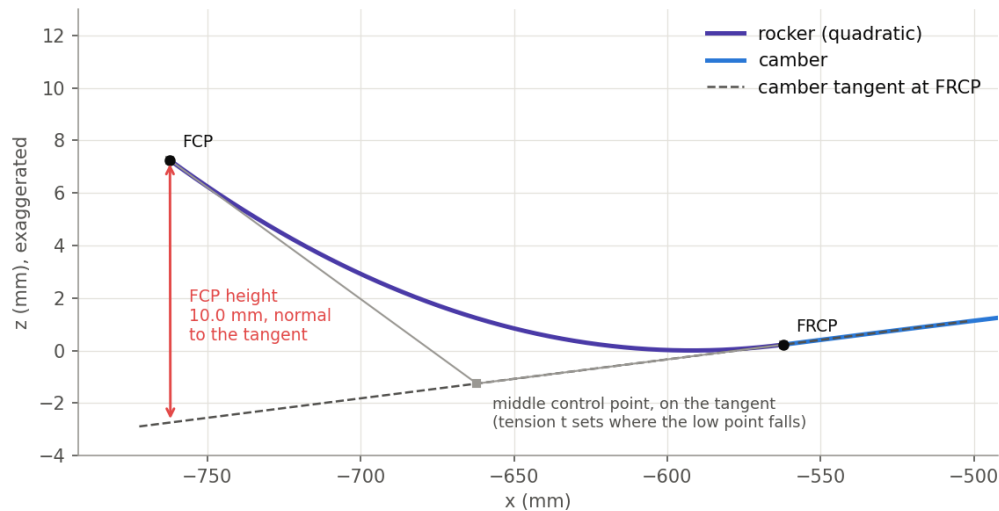
- Start a new ski's side profile
- Try camber / rocker targets quickly
- Shape the camber from the real EI



Test studio: NewTestBaseline, three edges (Z exaggerated 60 x).



Camber and rocker



Camber pocket

Pinned at FRCP and ARCP. With EI: the bending shape under a load at the mount. Without: a cubic peaking at the mount.

Rocker

A quadratic leaving FRCP tangent to the camber, ending FCP height below that tangent (measured normal to it).

Levelled

Rotated about the mount so both low points share Z; camber re-measured and corrected to 0.001 mm.



The dialog (1/2)

Generate baseline 1 ✓ ✗

1 SINGLE_CURVE CURVE_PER_REGION

2 Output curve name NewTestBase

3 FCP
Vertex of Derived 1 × 🕒 (Q)

4 ACP
Vertex of Derived 1 × 🕒 (Q)

5 Mount / load point
Vertex of Origin × 🕒 (Q)

6 ☒ Baseline targets

Camber height (MCH) 5 mm

Forebody rocker length 200...

☐ Spec forebody minimum

Aftbody rocker length 200 ...

☐ Spec aftbody minimum

FCP height 10 mm

ACP height 6 mm

#	Parameter	What it means
1	Output type	Curve per region (3 edges, one wire) or Single curve.
2	Output curve name	Default 'Baseline'.
3	FCP	Front contact point (X only).
4	ACP	Aft contact point (X only).
5	Mount / load point	Between FCP and ACP: where camber height is measured and the load acts.
6	Baseline targets	Camber height (0-15 mm), fore / aft rocker length, optional low-point distance, FCP / ACP height.



The dialog (2/2)

7 ☒ Use EI profile

EI profile edges (Q)

Edge of EI and Cross Sect... x

8 ☒ Spline output

Approximation tolerance 0....

Maximum control points 40

Curve degree 3

9 ☒ Add baseline sketch

10 ☐ Create weighted baseline

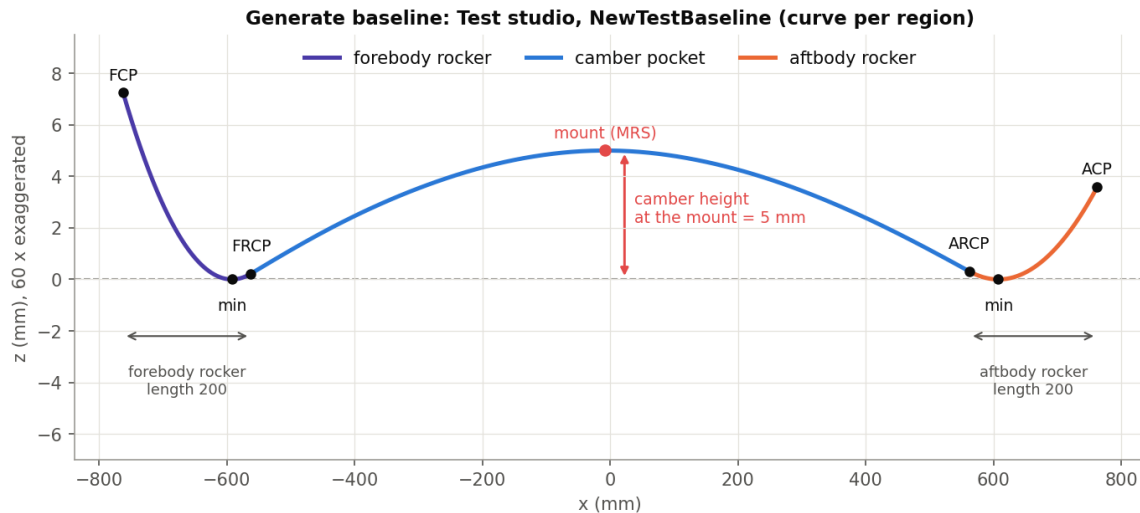
11 > Debug

Final

#	Parameter	What it means
7	Use EI profile	+ EI edges: camber = the ski's real bending shape.
8	Spline output	Tolerance, max control points, degree.
9	Add baseline sketch	The measurement sketch (as Analyze baseline draws).
10	Create weighted baseline	A second wire: camber flattened, rockers rotated with it.
11	Debug	Print setup, solver passes, analysis.



Example: NewTestBaseline



Camber 5 mm, rockers 200 / 200, FCP height 10, ACP height 6, Use EI profile on

Camber exactly 5.0 mm at the mount ($x = 0$). The tips end 7.2 and 3.6 mm above the low points: tip height is measured from the camber tangent, not from $Z = 0$.



Outputs and messages

Outputs

Key	What it holds
<Output curve name>	The baseline wire (camber, fore rocker, aft rocker; or one spline).
Baseline (Weighted)	Optional second wire.

Messages

	Message / what to do
Info	minimum-point distance clamped The requested rocker low point cannot be reached; the nearest reachable is used.
Error	mount not between FCP and ACP



Tips and limits



Two camber heights

Generate baseline sets it at the mount; Analyze baseline reports the largest perpendicular distance -- they differ with an off-centre mount.



No convergence notice

If 20 correction passes do not reach 0.001 mm, nothing is reported.



Where it lives

Document	xSection (f8deedeb1fbd819a8fa20113), 'Stiffness Tools'
Tab	generateBaseline (+ beam/generateBaselineSolver.fs, baselineCore.fs)
Tests	none
Examples	Test studio: Generate baseline 1
Explainer	docs/explainers/xsection/xsection_explained.md, section 2.5
Related	El and Cross Section, Estimate Stiffness, Estimate Deflection, Generate / Analyze baseline, Update profile